Omar Alfredo Castaño

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LinkedIn | CV (Spanish version) | CV (English version)

PROFESSIONAL SUMMARY

Senior Electronics Engineer with over 40 years of experience in Telecommunications and Spectrum Engineering, with a distinguished career in key Argentine regulatory bodies (ENACOM and its predecessors AFTIC, CNC, and SECom) and the private sector.

Expert in planning, assignment, coordination, and management of the radio spectrum, as well as in the design and deployment of terrestrial and satellite radio networks in geostationary (GSO) and non-geostationary (non-GSO) orbit.

Technical leader in complex telecommunications projects for major private companies and the Argentine Digital Terrestrial Television System (SATVD-T); specialist in the full lifecycle of low, medium, and high-power antennas and radiant systems.

Currently, I direct my expertise toward the architecture and decisive application of Generative Artificial Intelligence (GenAI) systems to optimize technical and creative processes in engineering and educational environments. This includes the development of **CognitiveOS** -an AI-assisted learning system based on controlled reasoning and knowledge distillation- aimed at integrating AI in a rigorous, verifiable, and useful manner within high-cognitive-demand educational environments, while always preserving institutional essence.

KEY TECHNICAL SKILLS

- ✓ **Spectrum Engineering**
- ✓ **Antennas and Radiating Systems**
- ✓ **GIS Software and Tools, Geospatial Data**
- ✓ **Generative Artificial Intelligence (GenAI) Systems**
- ✓ **Radiocommunications**
- ✓ **Technical Management and Consulting**
- ✓ **Regulations and Standards**
- ✓ **Cognitive Systems Architect (CognitiveOS)**

PROFESSIONAL EXPERIENCE

Consultant in Innovation and Generative AI (GenAI) Ecosystems — 09/2025 – Present

- **CognitiveOS (Architect of the AI-Driven Cognitive Educational System):**
  - **Concept:** Institutional reasoning and memory platform designed to mitigate Large Language Model (LLM) hallucinations.
  - **Innovation:** Implementation of a semantic distillation pipeline linking academic and audiovisual sources with direct traceability to the original source.
  - **Impact:** Transformation of scattered data repositories into auditable knowledge assets, preserving the institution's pedagogical identity.
  - **Status:** In architecture validation and prototyping phase, open to strategic partnerships.
- **Workflows with LLMs:** Gemini (Google), GPT (OpenAI), Claude (Anthropic), Grok (xAI), Meta AI, DeepSeek, for precision technical analysis and generation of adaptive regulatory/educational content, and personalized tutoring.
- **Google 2026 Development and Experimentation Ecosystem:**
  - **AI Models:** Gemini 3.1 Pro/Flash (with Deep Think logical reasoning for scientific tasks), Nano Banana Pro (images and infographics), Veo 3.1 (4K cinematic video) y Lyria 3 (music composition).
  - **Development Platforms:** AI Studio and Vertex AI for scaling; Antigravity as an agentic IDE with native n8n integration (MCP).
  - **Creative Tools:** Stitch (UI/UX design), Whisk (visual iteration), Flow (generative film production), and Google Vids (visual storytelling for enterprises).
  - **Productivity and Research:** NotebookLM for document synthesis and Video Overviews (analysis and summarization of YouTube) all in the cloud, and SurfSense (OSS and local LLMs) for knowledge extraction from private and local technical repositories.
- **Structured External Memory Systems:** Obsidian, SiYuan Note.
- **Multimodal Production and Streaming:**
  - **Video/Audio Editing and Post-production:** CapCut y DaVinci Resolve.
  - **Live Streaming and Video Recording:** OBS Studio (with AI via plugins/add-ons).
  - **Media Generation:** Music: Suno, Lyria 3 (Google), and Mureka; Audio, image, and video: ElevenLabs Studio, Sora 2 (OpenAI), Nano Banana 2 and Veo 3.1 (Google), Grok Imagine 1.0 (xAI), Movie Gen (Meta), Runway, Midjourney, LTX Studio, Seedance 2.0, Kling 3.0, MiniMax / Hailuo 2.3, WAN 2.6, and other AI video generation tools.

Senior Professional - Satellite and Open Digital Television (TDA) Technical Support, Directorate of Planning and Convergence (DINAPYC), National Communications Agency (ENACOM)) **07/2015 – 08/2025**

- **Regulatory technical leadership:** Led the continuous development of advanced methodologies, software applications, and regulatory frameworks for GSO/non-GSO satellite radiocommunication services and terrestrial fixed PtP/PtMP, mobile, and broadcasting services.
- **Updated key technical regulations:**
  - **GMC N° 64/97:** Procedures Manual for Coordination between Earth Stations (GSO and non-GSO) and Terrestrial Fixed Service Stations within MERCOSUR countries (100 MHz - 174.8 GHz).
  - **General Directive DG 61-03:** Interference calculation in Fixed Services in the range 0.03 - 174.8 GHz.
  - **FM Radio regulatory framework (76 - 108 MHz)** aligned with MERCOSUR GMC N° 47/2022 del MERCOSUR.
- **Developed advanced and innovative methodologies, software tools, and technical reports:**
  - **Electromagnetic Compatibility (EMC) between Terrestrial Fixed Service (PtP DFRS) and non-GSO Earth Stations (ES) of the FSS (e.g., Gen2 Starlink Gateway Site) in E-band (71 - 76 GHz (downlink) y 81 - 86 GHz (uplink))**, including a non-GSO satellite orbit predictor (e.g., Starlink Gen2 constellation) and calculation of coordination zones around the non-GSO FSS Earth station.
  - **3D radiating system measurement (30 MHz - 44 GHz) using unmanned aerial vehicles (RPA/UAV)**, including planning and execution of flight plans with supervised automatic operations (VLOS/BVLOS), reducing measurement campaign time and ensuring regulatory compliance (ANAC Resolution Nº 550/2025 and RAAC Part 100).
  - **Coverage and signal level calculation (wanted/interfering) with outdoor and indoor reception, for terrestrial area services in the frequency range 30 MHz to 6 GHz**, based on ITU-R Recs. P.1546 and P.1812.
  - **Prediction of wanted and interfering signals (co-channel/adjacent) in UHF SFN networks (470 - 698 MHz, Channels 14 - 51) for DTT coverage calculation (ISDB-T/TB) in Argentina's Open Digital Television (TDA).**
  - **Assessment of the impact of directional receiving antennas in DTT UHF MFN networks.**
  - **Specifications for UHF DTT receiving antennas.**
  - **Specifications for a Single High-Power Antenna System for FM Radio and VHF/UHF DTT transmission.**
  - **Analysis of the LeoLabs Inc. S-band PAFR (Phased Array Offset-Fed Reflector) Space Radar System** for detection and tracking of objects and space debris in LEO, assessing its impact on Space Situational Awareness (SSA) and Space Traffic Management (STM). Includes real case studies of satellite tracking (e.g., SAOCOM 1A and STARLINK-1347).
  - **Comprehensive planning of MVDDS services (12.2 - 12.7 GHz):** Channelization, development of coverage tools, and interference mitigation with DBS/DIRECTV. Includes on-site technical supervision of the MM Comunicaciones S.A. network.

Engineering Coordinator of the Advisory Council of the Argentine Digital Terrestrial Television System (SATVD-T) Ministry of Planning, Public Investment and Services (MinPLAN) **03/2010 – 06/2015**

- **Led the engineering coordination of SATVD-T**, from technology assessment to network optimization.
- **Managed DTT (Digital Terrestrial Television) network planning** in multi-frequency (MFN), single-frequency (SFN), and mixed MFN-SFN topologies, including infrastructure design, UHF spectrum optimization, and system administration.
- **Supervised the generation of over 2300 technical documents:** reports, technical data sheets for DTT Stations (EDT), simulations, GIS maps, resolutions, and key technical standards for SATVD-T (NT 001-004 and RE 003).
- **Designed and specified more than 240 DTT radiating systems:** Types A, B, C, D, and D+ for Medium and Low Power (DTT-FM) and Differential Type for High Power (DTT-FM) for the sites: Ministry of Public Works Building (MOP), Alas Building (Alas), Devoto, Florencio Varela, and the Avellaneda Single Communications Tower (Avellaneda (TUC).
- **Performed over 2900 coverage/interference simulations, generated over 2900 thematic GIS/SDI maps, and executed NIR (far-field) calculations for all SATVD-T station types.**
- **Designed and developed the link budget calculation methodology, along with a software application tool, for verifying Quality of Service (QoS) and availability of SDH 2 + 1 STM-1 (155.52 Mbit/s) Studio-Transmitter Links (STL)** with combined space and frequency diversity (6-8 GHz / 11-13 GHz).
- **Conducted interference analysis for SATVD-T and the Federal Authority for Audiovisual Communication Services AFSCA).**
- **Planned network optimization strategies (Phase IV), including the development of DSM/Clutter models and advanced propagation algorithms.**
- **Led strategic international technical commissions for the design of the Avellaneda Single Communications Tower (TUC)**, assessing technologies and best practices through visits to top-tier manufacturers, specialized laboratories, antenna measurement fields, and emblematic transmission towers worldwide:
  - **TECHNICAL MISSION** (04/06/2013 – 15/06/2013):
    - **AUSTRALIA:** RFS (Radio Frequency Systems) and Towers at TXA (TX Australia Pty Limited) transmission sites Ornata Road and Eyre Road.
  - **TECHNICAL MISSION** (25/06/2010 – 09/07/2010):
    - **BRAZIL:** KATHREIN (Kathrein Broadcast GmbH), São Paulo Tower.
    - **GERMANY:** LS Telcom (LS telcom AG), R&S (Rohde & Schwarz GmbH), SPINNER (Spinner GmbH), KATHREIN, Wendelstein Mountain Tower, Munich Single Tower.
    - **SPAIN:** RYMSA (currently Sener), MIER (Mier Comunicaciones S.A.), Torrespaña Radiotelevisión Española (RTVE), Collserola Tower.
- **Visit to the CPqD Foundation:** Evaluation of Service Provision and Tool for Spectrum Control in Argentina Related to Broadcasting Services (AM and FM) and Audio and Image (TV), for AFSCA (11/11/2013 al 14/11/2013).

- **Factory visit to CONSULFEM S.A. and the Antenna Measurement Laboratory (LaMA) of the National Space Activities Commission (CONAE):** Conducted the world's first validation of UHF DTT radiating systems with slant polarization, an innovation I proposed and validated (24/06/2013).
- **Provided expert technical advice and design for the initial deployment of the National Digital Terrestrial Television Systems in Venezuela (SVTVD-T), Bolivia (SBTVD-T), and Paraguay (SPTVD-T).**
- **Technical Responsible for the Commissions for the Initial Deployment of the SATVD-T (Argentina) and SBTVD-T (Bolivia) Systems.** Led the process of studying, site selection, and coverage calculation for the installation of High and Medium Power Digital Terrestrial Television (DTT) Transmitting Stations (similar to the EDTs used in SATVD-T).
  - **Bolivian Commission (Initial Phase of SBTVD-T)** (03/06/2013 – 17/06/2013).
  - **Argentine Commission (Initial Phase of SATVD-T** (01/03/2010 – 30/09/2010).

CNC Technical Representative and Member of the Coordinating Committee for the Specific Cooperation Project CNC-CONAE (PEC CNC-CONAE) — 10/2005 – 01/2009

- Managed CNC-CONAE technical collaboration for Electromagnetic Compatibility (EMC) between Earth Exploration-Satellite Service systems and the terrestrial Fixed Service.
- Coordinated the exchange/integration of technical products (orbital prediction software, GIS, etc.).
- Developed interference simulation software tools for EMC analysis in shared bands (FS vs. Earth Stations operating with Earth Exploration Satellites (EESS)) at CONAE's Córdoba Earth Station (located at the Teófilo Tabanera Space Center (CETT), in Falda del Carmen) which operates with the Argentine satellites SAOCOM and SAC-A/B/C/D/Aquarius, and international satellites such as Landsat, Spot, EROS, Terra, Aqua, NPP, NOAA, and GOES.

Coordinator of the Spectrum Engineering Work Team (ETIE) — 10/2004 – 01/2009

- Led a multidisciplinary team for the modernization of regulations and Technical Analysis tools.
- Development of methodologies and tools for spectrum engineering and management.
- Developed 57 Technical Reports (ITs), 27 Draft Technical Standards (PNT), 11 Functional Specifications (EFs) for software, and the 7 volumes of the Radio Spectrum Engineering Manual (Manual IER).
- Specified Digital Terrain Models (DEM/DSM) and the CNC GIS.
- Analyzed GSO/non-GSO Satellite Systems.

Coordinator of Spectrum Engineering Affairs, CNC Engineering Management — 05/2000 – 09/2004
Co-author of the Integrated Administration System (SIA) and Technical Analysis Projects — 01/2004 – 09/2004

- **Updated and developed new General and Technical Directives (DGs/DTs),** to harmonize technical criteria of the CNC Engineering Management.
- **Developed the technical standard:** Digitization of envelopes of Terrestrial Antenna Radiation Patterns operating at frequencies above 30 MHz to 60 GHz, presented at MERCOSUR (Montevideo 31/07/2001), CICOMRA (06/09/2001), and COPITEC (19/12/2001).
- **Author of the technical project:** Expansion of the Coverage Area of the Cellular Mobile Radiocommunication Service (SRMC) for the City and the International Airport of El Calafate, the Perito Moreno Glacier, and Access Routes, carried out April 21-24, 2000.
- **Co-author of General Directives (DG 33, DG 36, DG 54, and DG 61) for the Fixed and Mobile Services.**

Head of the Non-Geostationary (Non-GSO) Sub-sector of the Space Services Area, CNC Engineering Management — 03/1997 – 09/2004

- Performed technical analysis, coordination, approval, and assignment of GSO and non-GSO space systems (Little LEO, Big LEO -MEO-, and Broadband LEO).
- Analyzed spectrum coexistence in the Teledesic LEO satellite system (non-GSO FSS), the network architecture (288 Broadband LEO satellites) to provide "Internet-in-the-Sky" services, Ka-band spectrum sharing from 18.8 to 19.3 GHz (space-to-Earth), and interference with Fixed Services (FS); concluding that coexistence is difficult and highlighting the Argentine decision (Resolution 1608 SC/98) to allocate the band exclusively to non-GSO FSS to prevent interference.
- Represented CNC as a technical expert at the ITU Working Party 4A (WP 4A) meeting in Geneva, where technical criteria for Ka-band 18.8-19.3 GHz sharing by non-GSO FSS systems were evaluated (18/03/2003 - 18/03/2003).
- Gave the presentation "Synthesis of Aspects Related to Space Services" at the Buenos Aires Institute of Technology (ITBA) on 19/09/2002.
- Integrated the commission responsible for drafting the Frequency Coordination Manual for Earth and Terrestrial Stations (in C and Ku Bands) of MERCOSUR Working Group N° 1 "Communications" (GMC Resolution N° 60/01) (1997 – 2001).
- Responsible for coordination of pioneering non-GSO earth stations:
  - **Orbcomm (LEOTELCOM-1, Little LEO type)** in the town of Justo Daract, San Luis Province.
  - **Globalstar (HIBLEO-4FL, Big LEO type)** in the town of Bosque Alegre, Córdoba Province.
- Responsible for international coordination of the ICO Global Communications satellite network (non-geostationary intermediate circular orbit system, ICO) with Terrestrial Fixed Service Systems in the shared downlink band (space-to-Earth) 2185 - 2200 MHz.
- Responsible for Validation of the FS/MSS Interference Simulation Program Version 2.00 for the international coordination of the ICO Global Communications satellite network with Terrestrial Fixed Service Systems, carried out at CPqD in Brazil.
- Participated in MERCOSUR commissions, and in the evaluation of the AFMS Computer System (Automated Frequency Management System), and international delegations as an expert in non-GSO Satellite Systems (ITU-R, WRC-2000, CITEL).
- Acting Head of the Space Services Area (1999).

Operations Manager (ENTESA) / Engineering and Development Manager (PCSA) **03/1994 – 02/1997**
New Technologies Company Inc. (ENTESA) **07/1994 – 02/1997**
Professional Communications Inc. (PCSA) (https://professionalcommunicationssa.com/) **03/1994 – 06/1994**

- Led multidisciplinary teams in comprehensive consulting and advanced technological projects for key clients:
  - **EDEER S.A.:** Design and implementation of the Corporate Communications Plan.
  - **Gas Natural BAN S.A.:** Design and implementation of the Comprehensive Communications Plan (Data, Voice, SCADA).
  - **Petrolera Argentina San Jorge S.A.:** Design and implementation of telecommunications systems for SCADA in oil fields.
  - **Argencard S.A.:** Design and implementation of a data radio link.
- Designed and implemented telecommunications solutions (SCADA, data networks, VHF/UHF mobile, Microwave) and managed end-to-end projects and tenders.
- Research and Development (R&D) of Special Radiating Systems with innovative technology (applied in projects for Gas Natural BAN S.A. and EDEER S.A.), precursors to later designs (Single Antenna System for FM Radio and VHF/UHF DTT), carried out by the SATVD-T Advisory Council between March 2010 and June 2015.

Head of Antenna Laboratory / Factory Manager (Acting) **07/1986 – 11/1993**
Antenas Profesionales S.A. **07/1991 – 11/1993**
AHF Antenas S.A. **07/1987 – 06/1991**
AMPO S.A.C.I.F.A. **07/1986 – 03/1987**

- Managed the complete lifecycle of antennas and radiating systems at three leading companies, including design, research and development (R&D), production, quality control, and testing.
- Designed and developed (CAD drawings) antennas and radiating systems for VHF TV (channels 2 - 13) and UHF (channels 14 - 83); FM radio (76 - 108 MHz) for low, medium, and high power (vertical, horizontal, and circular polarization); and terrestrial radiocommunications for fixed and base stations in VHF and UHF, including omnidirectional and mobile antennas (UHF Double 5/8 Wave, 800-900 MHz cellular line), and directional antennas (Yagi up to 13 elements, with flat grid reflectors, dihedral, and parabolic up to 4 m in diameter)
- Designed RF Components for VHF and UHF for low, medium, and high power (broadband harnesses (up to 26 ways), adjustable high-power dividers (up to 8 ways), harmonic filters, and baluns.
- Designed and implemented two VHF/UHF antenna measurement fields and defined the associated measurement methodologies and protocols.

Specialist Engineer (05/1983 – 06/1986) / Technician (04/1980 – 04/1983) - Assignments **04/1980 – 06/1986**
Department, National Directorate of Radiocommunications, Secretariat of State for Communications (SECom)

- Progressed from Technician to Specialist Engineer at SECom, supervising analysis, approval, and assignment of frequencies/systems for Fixed Service (single-channel and multi-channel in VHF/UHF/Microwave) and Mobile Service (single-channel in VHF/UHF) under national and international regulations (ITU).
- Developed prediction methodologies for wanted and interfering signals for quality calculation in Fixed Service (FS) radiocommunication systems, operating at frequencies above 30 MHz within the VHF, UHF, and microwave bands. This work was carried out in a joint working commission between SECom (Spectrum Engineering and Assignments Departments) and the Argentine Chamber of Electronic Industries (CADIE), with participation from its main associated companies (TELETTRA ARGENTINA S.A.I.C., THOMSON-CSF ARGENTINA S.A.I.C., GTE INTERNATIONAL INC., and EASTEL S.A.I.C.).
- Acting head of the VHF, UHF, and SHF (Microwave) Section (1985).

RELEVANT CONTINUING EDUCATION (Selection)

- Self-directed continuous learning in Management and Application of GenAI Systems. — 2024 – Present
- ITU Seminars / International Forums (Radiocommunications, VSAT, GMPCS, Satellites). — 1997-2008
- .NET Development Courses. — 2005
- Seminars on Space Systems and Interference. — 1998 y 2008
- Courses on Satellite Link Calculation and Satellite Communications. — 1997-1998
- Specialized Technical Courses: RF Measurements, TV Systems, and Communications. — 1983-1986

EDUCATION

- **Postgraduate Degree in Digital TV - University of Palermo.** — **2009-2011**
- **Electronic Engineer - Avellaneda Regional Faculty of the National Technological University (FRA-UTN).** — **1978-1984**
- **Electronics Technician (Telecommunications) - ENET N° 3 "Dr. Salvador Debenedetti".** — **1970-1976**

HABILIDADES TÉCNICAS CLAVE

- **Spectrum Engineering:**
  - Spectrum Planning, Allocation, Management, and Optimization.
  - National and International Coordination (MERCOSUR, CITEL, ITU).
  - Interference and EMC Analysis (terrestrial services, space services (GSO/non-GSO), and mixed scenarios (E-s/s-E)
  - Advanced Radio Wave Propagation Modeling (ITU-R P Series Recommendations).
- **Radiocommunications:**
  - **Terrestrial Broadcasting:** FM Radio, Analog VHF-Lo/Hi TV, VHF/UHF DTT (ISDB-T/TB, DVB-T/T2).
  - **Satellite Systems:** GSO and Non-GSO/NGSO (SSO/LEO/MEO/HEO). Key satellite services: FSS, BSS, DBS, EES, MSS, RNSS, etc.

- **Fixed Terrestrial Systems:** Point-to-Point (PtP) and Point-to-Multipoint (PtMP) (VHF/UHF and Microwave (SHF/EHF)) including backhaul radio links for mobile networks (4G, 5G, and future 6G) high capacity in E Band.
- **Mobile Systems:** 3G to 5G NR (6G perspective).
- **Specialized Systems:** MVDDS, wireless SCADA, space radar (PAFR), ILS.

- **Antennas and Radiating Systems:**
  - Theory, design, simulation (2D/3D), development, prototyping, and testing (Lab/Field).
  - Measurement and technical verification (Fraunhofer far-field and Fresnel intermediate-field regions, RPA/UAV).
  - Characterization, Production, Tuning, Quality Control, and Maintenance.
  - **Types:** Parabolic, Dihedral Reflectors, Flat Panels, Yagi-Uda, Dipoles with/without Reflector, Mobile, Simple Arrays, and Radiating Systems (omnidirectional, sectorial, and directional), including High-Power Single Antenna Systems for FM Radio and UHF DTT transmission (with linear, elliptical, and/or circular polarization).
  - **Parameters:** Patterns, Gain, Directivity, Polarization (Linear, Elliptical, Circular), Down Tilt, Null Fill, Zin, VSWR, etc.
- **Technical Management and Consulting:**
  - Telecommunications Network Planning and Design (Fiber Optics, Radio Links (VHF/UHF/Microwave), Digital Trunked Radio System, Wireless SCADA System).
  - End-to-end technical project management.
  - Development of Technical Specifications and Tender Documents.
  - Technical-Economic Evaluation of Solutions and Supplier Selection.
  - Implementation, Operations, and Maintenance (O&M) Supervision.
- **Engineering and Management Software:**
  - **Simulation and Design:** MATLAB, Pathloss, Radio Mobile, ICS Telecom, AutoCAD, MicroStation, DesignCAD.
  - **Geospatial:** ArcGIS, MapInfo, ERDAS Imagine, Google Earth, and ITU Tools (IFIC, GIMS, SpaceCap, SPS).
  - **Development:** .NET, Python.
  - **Machine Learning:** PyTorch, TensorFlow (currently in research and self-learning).
  - **And other Generative AI ecosystem tools.**
- **GIS and Geospatial Data:**
  - **Digital Terrain Model (DTM):** SRTM 3/1 arc-second (Shuttle Radar Topography Mission 3/1 arc-second (approx. 90/30 m resolution at the equator)), understanding that native SRTM data is a Digital Surface Model (DSM) used as a DTM substitute, in raster format (e.g., GeoTIFF - tif).
  - **Digital Surface Model (DSM):** Includes clutter data (information on land cover types -e.g., urban, forest, suburban, rural, water- and the height of obstacles above the terrain, such as buildings and vegetation), in raster (e.g., GeoTIFF - tif) or vector (e.g., Shapefile - shp) formats.
  - **Building and Urban Fabric Cartography:** 3D buildings in vector format (e.g., Shapefile - shp).
  - **Projections:** Geographic (latitude/longitude), GK (Gauss-Krüger), UTM, UPS, and **Reference Systems:** WGS84, EGM96.
- **Regulations and Standards:**
  - **National Argentina (ENACOM):** General and Technical Directives (DGs and DTs) and Resolutions.
  - **International:** ITU Radio Regulations (RR), ITU-R/ITU-T Recommendations, MERCOSUR/CITEL Agreements, ETSI Standards, FCC, 3GPP, etc.
- **Other Competencies:**
  - **Verification of EMF exposure limits (ICNIRP) for non-ionizing radiation assessment at broadcasting, mobile, and fixed stations**, in accordance with current national regulations (MS Resolution 202/95, CNC Resolution 3690/04).
  - **Non-GSO satellite orbit predictor** using TLE according to ITU-R Rec. S.1503-3 and SGP4 (LEO) and SDP4 (MEO/HEO) models.
  - **Quality of Service (QoS) and Availability of Digital Point-to-Point Fixed Radio Systems (DFRS PtP)** (PDH 2/8/34/140 Mbps and SDH STM-1/STM-4/STM-16 networks) for frequencies up to 175 GHz and path lengths up to 60 km.